CS6501 Generative AI Project B

# CardioRAG-AGENT

## A Tool-Using, Retrieval-Grounded, Self-Correcting Multimodal Reasoning Agent

Video Demo: https://www.youtube.com/watch?v=0zYNCz7W3qw
Code Base: https://github.com/trankxu/uva-26-spring-gen-agent-students-projectB

**Team 6**

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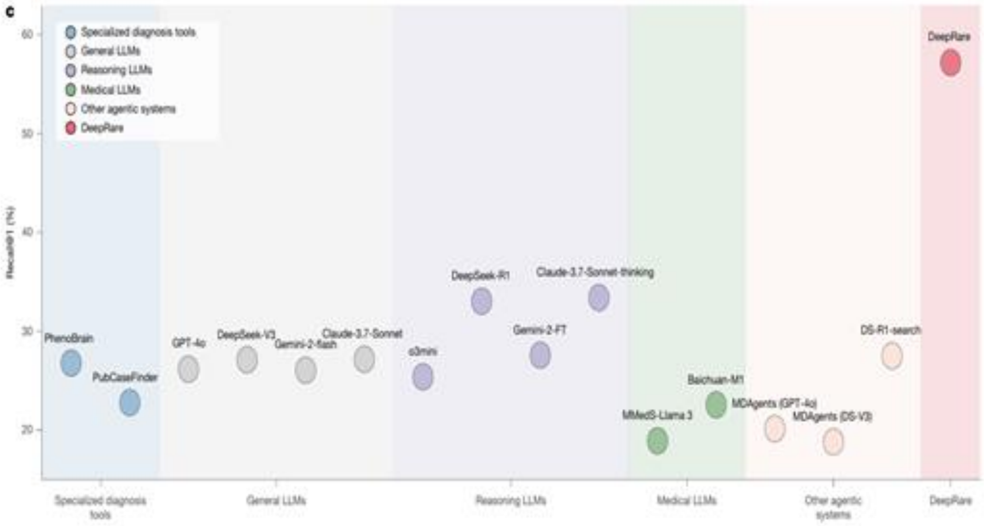
F April 28, 2026

# Motivation

- **Why Agent?**
  - Limited specialist availability
  - Rising clinical workload
  - Diagnostic odyssey



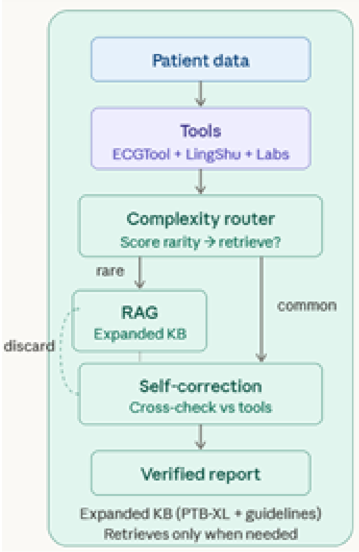

- **Current Medical Agent Gaps**

  The Modality Gap: Unimodal Ambiguity -> Multimodal Synergy
  The Precision Gap: Weak Numerical Rigor -> Measurement Tool Use
  The Grounded Gap: Ungrounded Hallucination -> Case-Based Retrieval (RAG)
  The Logic Gap: Physical Inconsistency -> Physics-Aware Self-Correction
  Challenges in Rare Diseases: Naive RAG fetches misleading clinical mimics

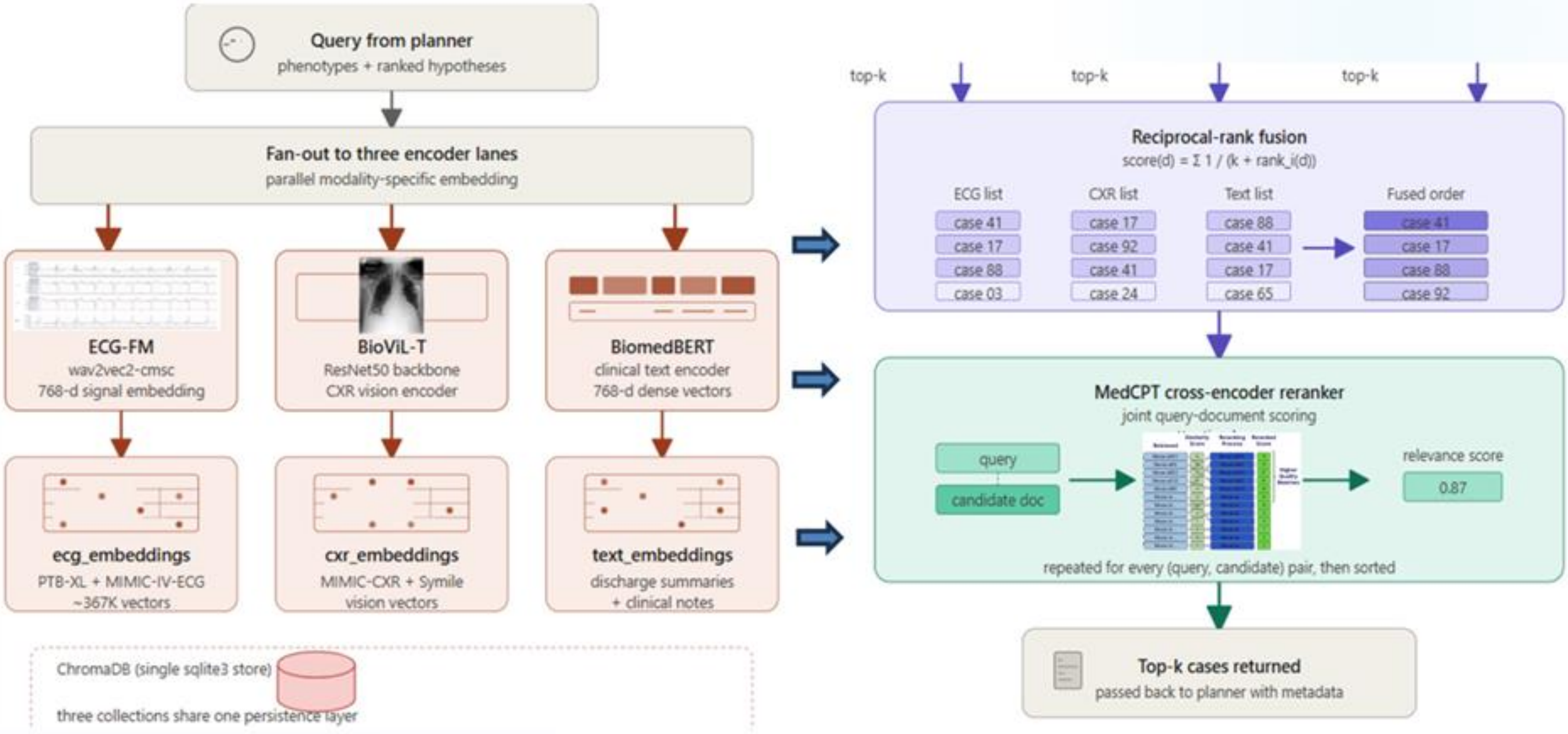
# Problem Definition: What to Work On?

Expand and improve RAG utilization by enriching the knowledge base with general medical knowledge and larger historical case archives;

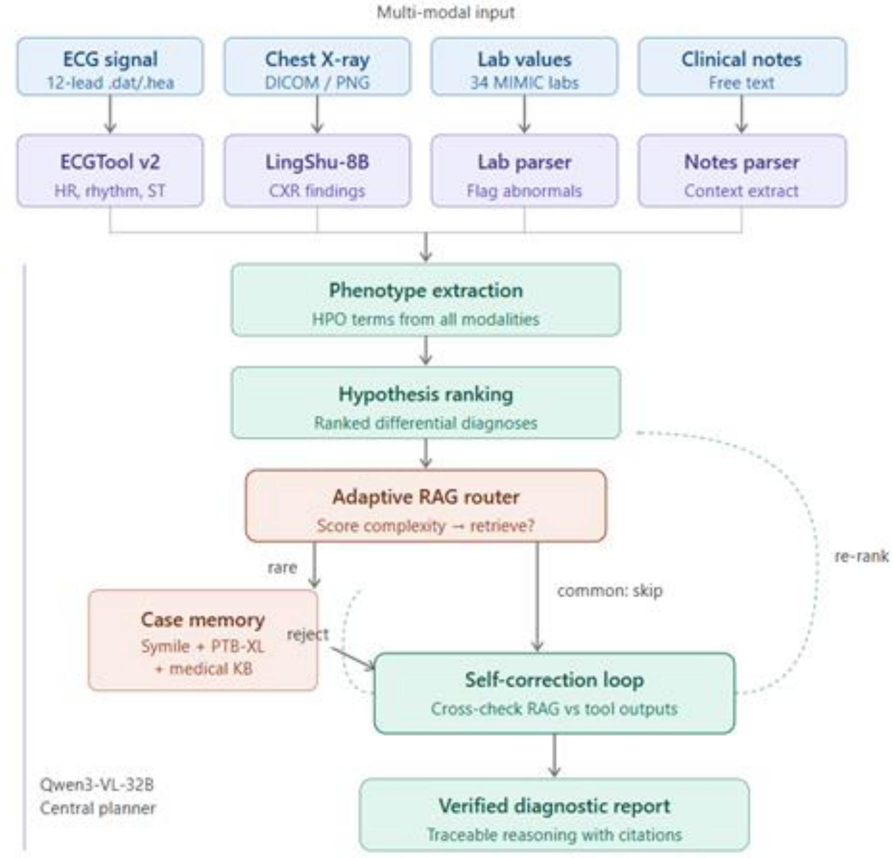
**Adopt the *DeepRare*-inspired architecture** to teach the Qwen Planner when to trigger retrieval, and how to critically evaluate retrieved cases against the patient's actual multi-modal findings before finalizing diagnosis.



# System Architecture



# System Architecture



# Method - Structured Reasoning

Inspired by DeepRare, Nature 2025

### 1. Phenotype Extraction

Free text → HPO terms 38 rules

"Cardiomegaly"
→ HP:0001640 [cxr]

"BNP 892 pg/mL ↑"
→ HP:0430025 [lab]

"ST elevation V1-V3"
→ HP:0012251 [ecg]

### 2. Constellation Match

14 clinical rules pattern-match phenotype sets

Cardiomegaly + Edema + ↑BNP
**→ CHF (0.85)**

ST elevation + ↑Troponin
**→ STEMI (0.85)**

Tachycardia + ↑D-dimer
**→ PE (0.50)**

### 3. Traceable Hypothesis

Each diagnosis carries its evidence chain

```
DiagnosticHypothesis
  diagnosis: CHF
  confidence: 0.84
  evidence: [
    cxr → cardiomegaly
    lab → BNP elevation
    cross_modal: agree
  ]
```

Every prediction can be answered with: which modality saw what, and how strongly.

# Method - Vector RAG

Multi-modal embedding retrieval — three specialized encoders fused via Reciprocal Rank Fusion

**Knowledge Base**: MIMIC-IV-ECG, MIMIC-IV-CXR, PTB-XL, Symile-MIMIC train set

**Advantages**: Keyword RAG misses synonyms — "heart failure" never matches "CHF", "MI", or "a-fib". Embeddings handle this natively.

### ECG

## ECG-FM

*12-lead signal*

Foundation model pre-trained on 1M+ ECGs.

### IMAGE

## BioViL-T

*Chest X-ray*

Biomedical vision-language transformer.

### TEXT

## BiomedBERT

*Notes + lab summary*

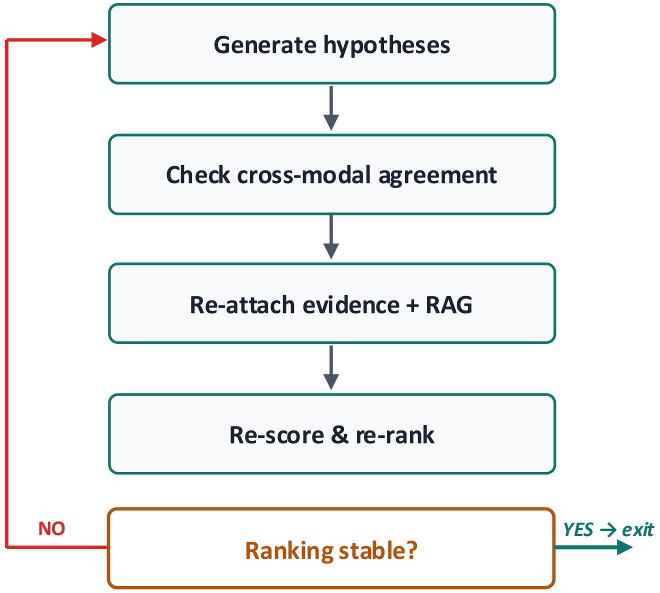
Domain-specific text encoder.

## RRF Fusion

$$\text{score}(c) = \Sigma \quad 1 / (k + \text{rank\_i}(c))$$

*sum across modalities → top-k cases*

# Method - Self-Reflection Loop




## Re-score formula

```
new = 0.50 × prior
    + 0.15 × support
    − 0.20 × contradict
    + min(0.10·n_modalities, 0.3)
    + min(0.10·xmodal_agree, 0.2)
    + min(0.05·rag_match, 0.15)
```

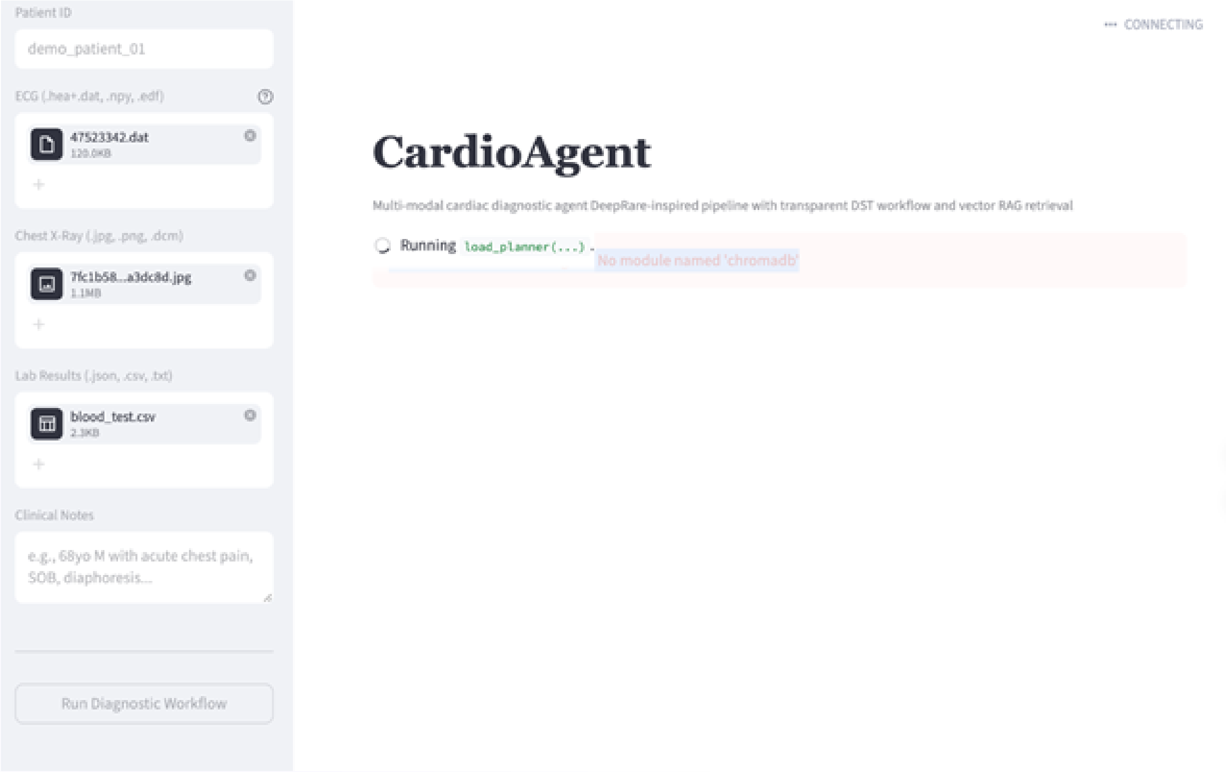
Contradictions weighted higher than supports.

Source weights

ECG/Lab 1.0 CXR 0.9 Notes 0.7 RAG 0.4

# Demo

End-to-end walkthrough on a Symile-MIMIC case



# Case Study: Diagnostic Trace (Sample 98)

**Setting A** Qwen-only

| | |
|---|---|
| Cardiomegaly | 0 ✓ |
| Edema | 0 ✗ missed |
| Atelectasis | 0 |
| Lung Opacity | 1 |
| No Finding | 0 |
| Pleural Effusion | 0 |

35.7s | Conservative — missed edema entirely

**Setting B** +Tools, no RAG

| | |
|---|---|
| Cardiomegaly | 1 ✗ false positive |
| Edema | 1 ✓ |
| Atelectasis | 1 |
| Lung Opacity | 1 |
| No Finding | 0 |
| Pleural Effusion | 1 |

84.9s | Aggressive — caught edema but over-predicted

**Setting C** +Tools +RAG

| | |
|---|---|
| Cardiomegaly | 1 ✗ false positive |
| Edema | 1 ✓ |
| Atelectasis | 1 |
| Lung Opacity | 1 |
| No Finding | 0 |
| Pleural Effusion | 1 |

64.6s | Same as B — RAG reinforced, didn't correct

| | | | |
|---|---|---|---|
| ✅ | Planning | 4 modalities detected (ECG, Image, Labs, Notes) | |
| ✅ | ECG | HR 88 bpm, normal sinus rhythm, 1 finding | 5.4s |
| ✅ | CXR (LingShu) | 1 finding — but connection error noted in report | 1.4s |
| ✅ | Labs | 34 values, **7 abnormal**: Cr↑ BUN↑ Lactate↑ WBC↑ Glucose↑ AST↑ ALP↑ | 0.0s |
| ✅ | Notes | 267 chars — mentions cardiomegaly, edema, pleural effusion | 0.0s |
| ✅ | Phenotypes | 10 HPO terms: Cardiomegaly, Pulmonary edema, Pleural effusion, Atelectasis... | |
| ✅ | Hypotheses | CHF (0.75), Pneumonia (0.55), Cardiorenal syndrome (0.55) | |
| ⏭ | RAG | Skipped — CaseMemory not loaded | |
| ✅ | Reflection | 2 rounds, 3 agreements, **0 contradictions** → Cardiorenal syndrome (1.00) | |
| ✅ | Synthesis | 5,352 chars — "Acute Decompensated Heart Failure with Cardiorenal Syndrome" | 78.1s |

Case 1: symile_train_24690376 → **Lung Opacity**
Case 2: symile_train_24494866 → **Atelectasis, Cardiomegaly, Edema, Lung Opacity, Pleural Effusion**
Case 3: symile_train_29481597 → **Atelectasis, Cardiomegaly, Lung Opacity, Pleural Effusion**
Case 4: symile_train_22072768 → **Atelectasis, Cardiomegaly, Edema, Lung Opacity, Pleural Effusion**
Case 5: symile_train_24217071 → **Cardiomegaly, Edema, Lung Opacity, Pleural Effusion**

4 of 5 retrieved cases have Cardiomegaly → RAG **reinforced** the false positive instead of correcting it.

# Experiment Setup

## Input Data Scope

**Evaluation: MIMIC-IV-ECG (ECG) + Symile-MIMIC (CXR/Labs), test split only**

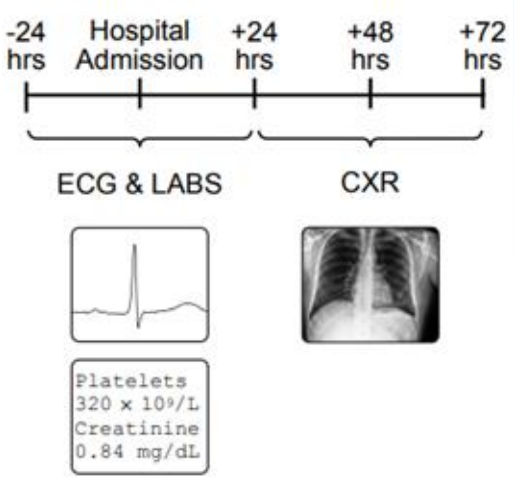
**External Knowledge: PTB-XL + MIMIC-IV-ECG + MIMIC-CXR-JPG, train split**

**ECG (Signal)**
Electrical Function

**CXR (Image)**
Anatomical Structure

**Doctor Notes (Text)**
Ground Truth Context

**Lab Results (Text)**
Ground Truth Context



# Evaluation - Ablation Study

Symile-MIMIC test set; MIMIC-IV-ECG test data

| Component | A | B | C | D |
|---|---|---|---|---|
| Qwen3-VL zero-shot | ✓ | ✓ | ✓ | ✓ |
| ECGTool + Lingshu imaging | — | ✓ | ✓ | ✓ |
| Phenotypes + Reflection | — | ✓ | ✓ | ✓ |
| Keyword RAG | — | — | ✓ | — |
| Vector RAG (RRF) | — | — | — | ✓ |
| Research question | *Baseline* | *Agentic?* | *Keyword?* | *Vector?* |

# Evaluation - Ablation Study

Symile-MIMIC test set; MIMIC-IV-ECG test data

| Setting | Description | n | Macro AUROC | Macro AUPRC | Macro F1 | Micro AUROC | Exact Match | Hamming ↓ |
|---|---|---|---|---|---|---|---|---|
| ***Block 1 — Pipeline ablation (200 samples)*** | | | | | | | | |
| A | **Qwen3-VL zero-shot** — CXR + labs only, no tools, no RAG | 187 / 200 | 0.4709 | 0.6441 | 0.5438 | 0.5307 | 0.2437 | 0.5340 |
| B | **DeepRare planner + ECG/CXR tools** — phenotype extraction + reflection, no RAG | 200 / 200 | 0.5443 | 0.6808 | 0.7144 | 0.5257 | 0.3163 | 0.4339 |
| C | **DeepRare + naïve text RAG (367K)** — keyword retrieval over MIMIC-IV-ECG | 200 / 200 | **0.5588** | **0.6883** | **0.7221** | **0.5304** | **0.3214** | 0.4355 |
| ***Block 2 — Vector RAG modality ablation (2,000 samples)*** | | | | | | | | |
| D1 | **ECG-FM only** — single-modality ECG embeddings | 2000 / 2000 | **0.5891** | **0.7101** | **0.7579** | **0.5518** | **0.3720** | **0.4058** |
| D2 | **BioViL-T (CXR) only** — single-modality CXR embeddings | 2000 / 2000 | 0.5867 | 0.7082 | 0.7563 | 0.5450 | 0.3654 | 0.4077 |
| D3 | **BiomedBERT (text) only** — single-modality clinical text embeddings | 2000 / 2000 | 0.5761 | 0.7058 | 0.7527 | 0.5451 | 0.3674 | 0.4095 |
| D4 | **All modalities + RRF fusion** — reciprocal-rank fusion across ECG / CXR / text | 2000 / 2000 | 0.5829 | 0.7073 | 0.7534 | 0.5463 | 0.3679 | 0.4111 |

# Takeaway & Future Work

- Medical Agent means traceable reasoning, not just better prompt.
- Full models （with vector RAG） work the best.
- For these CXR labels, ECG already carries most of the signal, so adding text and image rankings just adds noise

**Future Work**

- Refine the front-end interface
- Improve performance
- Add more rules

Thank you! Any Questions？